

HE2003 Econometrics I

Tutorial 7

14/10/2024

Question 1

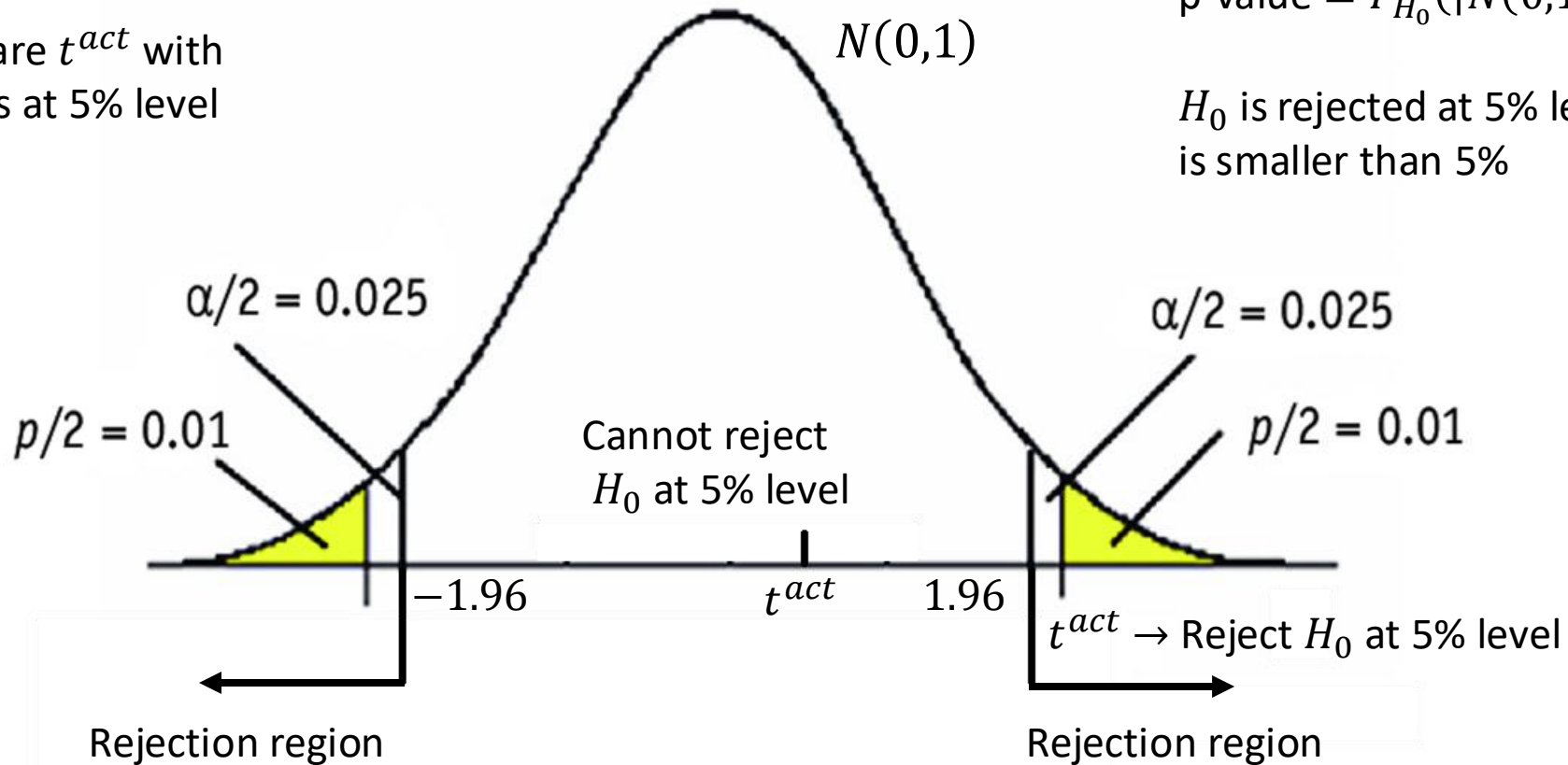
- (a) For testing $H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$ at the 5% significance level, compare the t-test procedure with the p-value procedure.
- (b) Explain the definition of a 95% confidence interval. Explain the trade-off between the 95% confidence interval and a confidence interval with a higher coverage probability such as the 99% confidence interval.
- (c) Explain the relationship between a t-test at the 5% significance level and a 95% confidence interval.
- (d) Explain the difference between homoskedasticity and heteroskedasticity.

Question 1 (a)

t-test: compare t^{act} with critical values at 5% level

$$p\text{-value} = P_{H_0}(|N(0,1)| > |t^{act}|)$$

H_0 is rejected at 5% level if p-value is smaller than 5%



Question 1 (b)

For the linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$, $i = 1, \dots, n$, the 95% confidence interval (CI) for β_1 collect all the possible values of β_1 that cannot be rejected by the corresponding (two-sided) t-test at the 5% significance level:

$$CI_{\beta_1, 95\%} = \left\{ \beta_1 : \left| \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \right| \leq 1.96 \right\} = \{ \beta_1 : |\hat{\beta}_1 - \beta_1| \leq 1.96 \times SE(\hat{\beta}_1) \}$$

Solving the inequality, we find that the 95% CI takes the following form:

$$CI_{\beta_1, 95\%} = [\hat{\beta}_1 - 1.96 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \times SE(\hat{\beta}_1)]$$

where $\hat{\beta}_1$ denotes the OLS estimator for β_1 and $SE(\hat{\beta}_1)$ denotes the standard error for $\hat{\beta}_1$. We can define the 95% CI for β_0 in the same fashion.

Note: the center of the CI is in fact equal to the OLS estimator $\hat{\beta}_1$ (that is, the CI is symmetric around the OLS estimator $\hat{\beta}_1$), and the length of the CI is equal to $2 \times 1.96 \times SE(\hat{\beta}_1)$. This implies that the smaller the standard error, the shorter the CI will be (smaller standard error means that the estimation for β_1 is more precise).

Question 1 (b)

To construct a 99% confidence interval, we just need to replace 1.96 with 2.58, which is the critical value for the (two-sided) t-test at the 1% significance level:

$$CI_{\beta_1, 99\%} = [\hat{\beta}_1 - 2.58 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 2.58 \times SE(\hat{\beta}_1)]$$

By definition, a 99% CI has a higher coverage probability than a 95% CI: it will cover the true value 99% of the time. However, the price to pay for getting a higher coverage probability is that the length of the 99% CI also has to be longer than that of the 95% CI.

Question 1 (c)

By the definition of a 95% confidence interval CI for β_1 , it collects all the possible values of β_1 that cannot be rejected by the corresponding (two-sided) t-test at the 5% significance level. Then, suppose the value 1 is inside the 95% CI for β_1 , then it cannot be rejected by the t-test at the 5% level. More specifically, we can conclude the result for testing $H_0: \beta_1 = 1$:

$$|t_1| = \left| \frac{\hat{\beta}_1 - 1}{SE(\hat{\beta}_1)} \right| \leq 1.96$$

where we use the notation t_1 to emphasize that we are testing $H_0: \beta_1 = 1$. On the other hand, suppose the value 2 is outside the 95% CI, then we can conclude the result for testing $H_0: \beta_1 = 2$:

$$|t_2| = \left| \frac{\hat{\beta}_1 - 2}{SE(\hat{\beta}_1)} \right| > 1.96$$

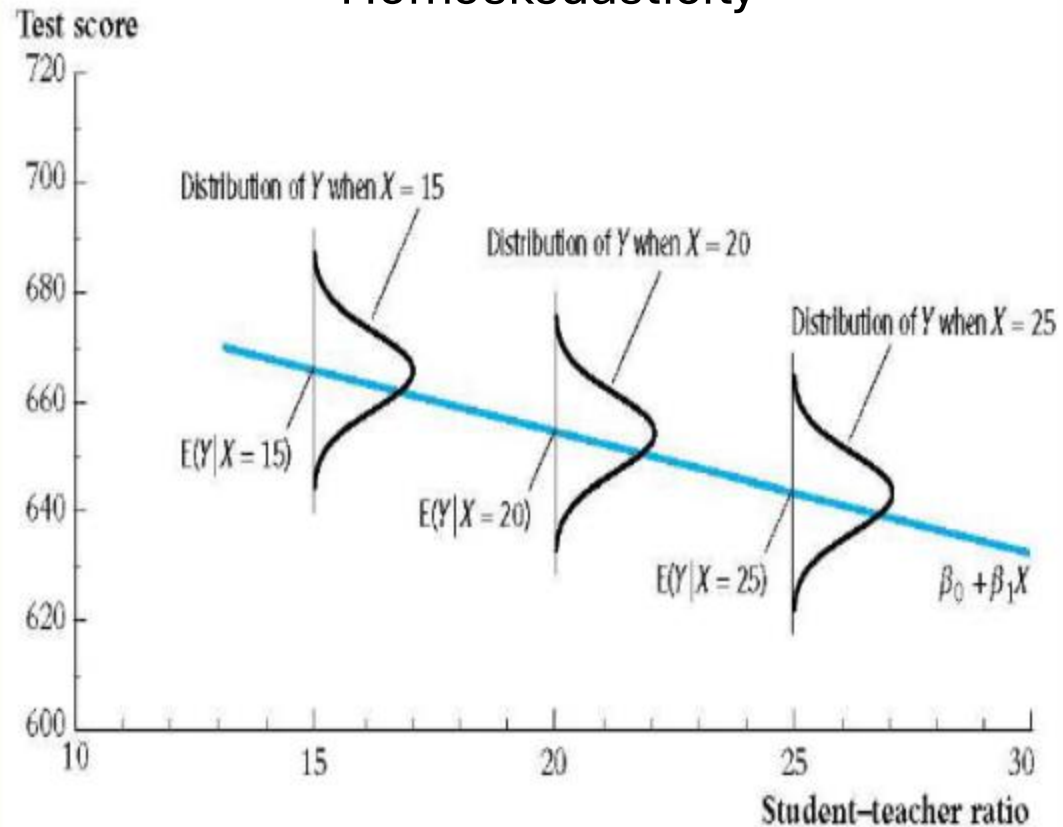
Question 1 (d)

Homoskedasticity: In the following figure, the distribution of Y (and the distribution of the errors u) is shown for different values of X (remember that the only difference between Y and u is the population regression line). All these distributions have the same spread: The variance of these distributions is the same for different values of X . Thus, the errors u in this figure are homoskedastic (the variance of u does not change with the value of X).

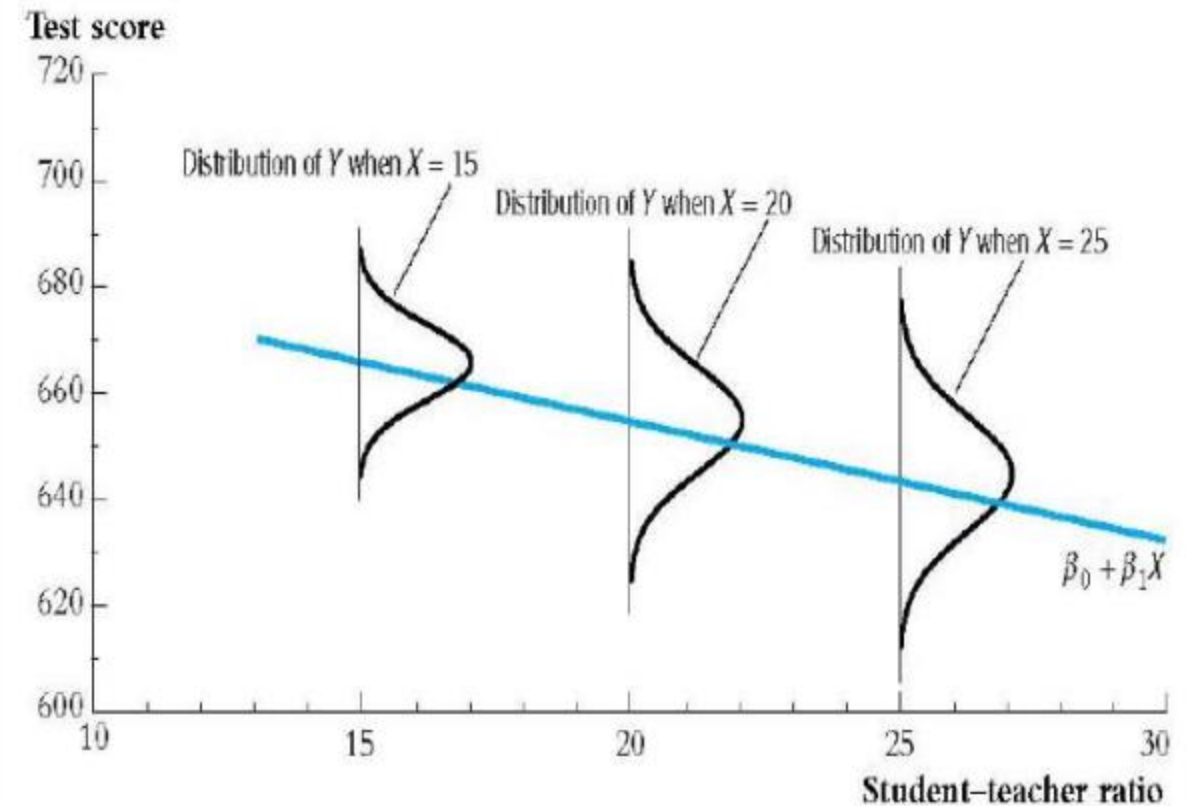
Heteroskedasticity: By contrast, the following figure illustrates a case in which the distribution of Y (and the distribution of u) spreads out as the value of X increases. In general, the errors u are called heteroskedastic when the variance of u changes with the value of X .

Question 1 (d)

Homoskedasticity



Heteroskedasticity



Question 2

Tennessee conducted an experiment in which kindergarten students were randomly assigned to “regular” and “small” classes and given standardized tests at the end of the year. Regular classes contained approximately 24 students, and small classes contained approximately 15 students. Let *SmallClass* denote a binary variable equal to 1 if the student is assigned to a small class and equal to 0 otherwise. A regression of *TestScore* on *SmallClass* yields

$$\begin{array}{ccc} \textit{TestScore} = 918.0 + 13.9 \times \textit{SmallClass}, & R^2 = 0.01 \\ (1.6) & (2.5) \end{array}$$

- (a) Do small classes improve test scores? By how much?
- (b) Is the estimated effect of class size on test scores statistically significant? Carry out a test at the 5% level.
- (c) Construct a 95% confidence interval for the effect of *SmallClass* on *TestScore*.

Question 2

(a) By using conditional expectation, we have:

$$\beta_1 = E[TestScore|SmallClass = 1] - E[TestScore|SmallClass = 0]$$

As the OLS estimate $\hat{\beta}_1 = 13.9$, on average small classes improve test scores by 13.9 (points).

(b) To test if the estimated effect of class size on test score is statistically significant, we compute:

$$|t| = \left| \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \right| = \left| \frac{13.9}{2.5} \right| = 5.56 > 1.96$$

Therefore, we can reject $H_0: \beta_1 = 0$ at the 5% level, which means that the estimated effect is statistically significant at the 5% level.

(c) By the definition of a 95% CI, we have:

$$\begin{aligned} CI_{\beta_1, 95\%} &= [\hat{\beta}_1 - 1.96 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \times SE(\hat{\beta}_1)] \\ &= [13.9 - 1.96 \times 2.5, 13.9 + 1.96 \times 2.5] = [9, 18.8] \end{aligned}$$

Question 3

For the linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$, a test of the null hypothesis $H_0: \beta_1 = 5$ using the t-statistic yields a p-value of 0.012.

(a) Does the 99% confidence interval contain $\beta_1 = 5$? Explain. Does the 95% confidence interval contain $\beta_1 = 5$? Explain.

(b) Suppose that the true value of β_1 is actually 10 and the standard error of $\hat{\beta}_1$ is equal to 1.

Can you illustrate graphically the Type II error of the t-test with 5% significance level for $H_0: \beta_1 = 5$? Is the Type II error smaller than 50%? Explain. [Hint: (1) remember the Type II error is accepting H_0 when H_0 is wrong, (2) we can draw the distribution of the t-statistic when the true value of β_1 is actually 10, and then find out the area corresponding to the Type II error.]

Question 3 (a)

For the null $H_0: \beta_1 = 5$, the p-value is equal to 0.012, which is larger than 0.01, so we cannot reject $H_0: \beta_1 = 5$ at the 1% significance level. Since we cannot reject it at the 1% level, by the relationship between t-test and confidence interval, we can conclude that the value $\beta_1 = 5$ will be contained in the 99% confidence interval.

Furthermore, since the p-value is smaller than 0.05, so we can reject $H_0: \beta_1 = 5$ at the 5% significance level. Since we can reject it at the 5% level, the value $\beta_1 = 5$ will not be contained in the 95% confidence interval.

Question 3 (b)

To find out the distribution of the t-statistic when the true value of β_1 is equal to 10 instead of 5, note that in general the following always holds under large samples (and when all the OLS assumptions are satisfied):

$$\frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim N(0,1)$$

Therefore, for testing $H_0: \beta_1 = 5$:

$$\begin{aligned} t_5 &= \frac{\hat{\beta}_1 - 5}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} + \frac{\beta_1 - 5}{SE(\hat{\beta}_1)} \\ \Rightarrow t_5 &\sim N(0,1) + \frac{10 - 5}{SE(\hat{\beta}_1)} = N(0,1) + 5 = N(5,1) \end{aligned}$$

which follows from the information that the true value $\beta_1 = 10$ and $SE(\hat{\beta}_1) = 1$. Therefore, when the true value of β_1 is 10, the distribution of the t-statistic for testing $H_0: \beta_1 = 5$ follows $N(5,1)$, a mean-shift toward right-hand-side by 5 from the $N(0,1)$ distribution.

Question 3 (b)

Then, we can draw the graph for the corresponding area of Type II error as follows. Remember the definition of the Type II error is: accepting H_0 when H_0 is wrong (when H_1 is correct), therefore we just need to find out the area corresponding to the H_1 distribution $N(5,1)$ that falls inside the Acceptance Region (from -1.96 to 1.96). Apparently, this area is smaller than 50% (see Figure 1). In general, remember that $N(0,1)$ is the distribution of the t-test when H_0 is true, $N(c, 1)$ with $c \neq 0$ is the distribution of the t-test when H_0 is wrong.

Similarly, if we consider the case where the H_1 distribution is $N(2,1)$, we see that now the Type II error becomes much larger as the distribution shifts toward left-hand-side, but it is still slightly smaller than 50%, as 2 is larger than 1.96 (see Figure 2).

Question 3 (b)

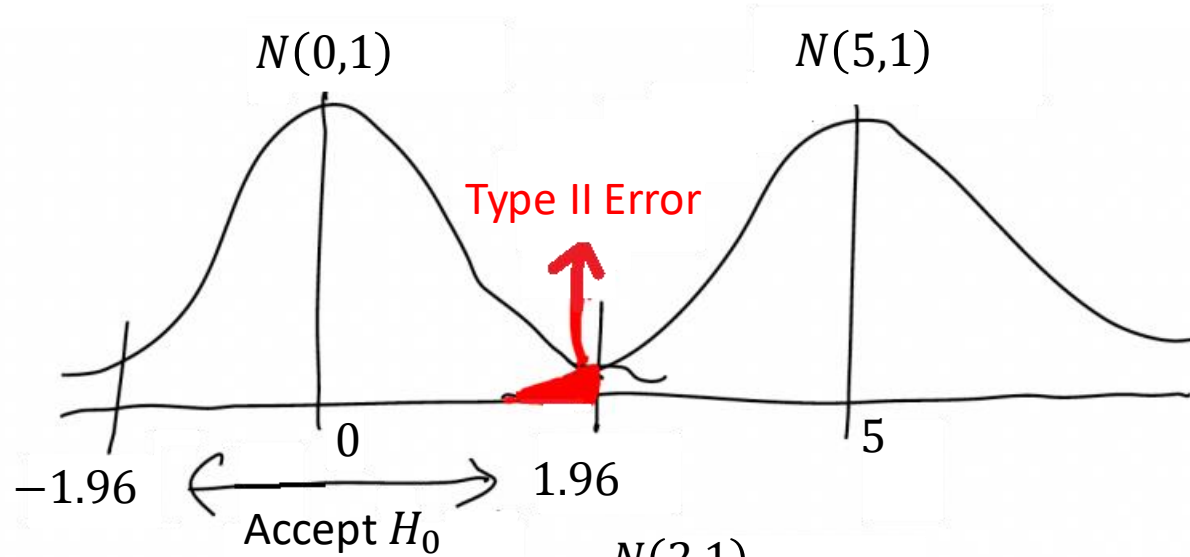


Figure 1

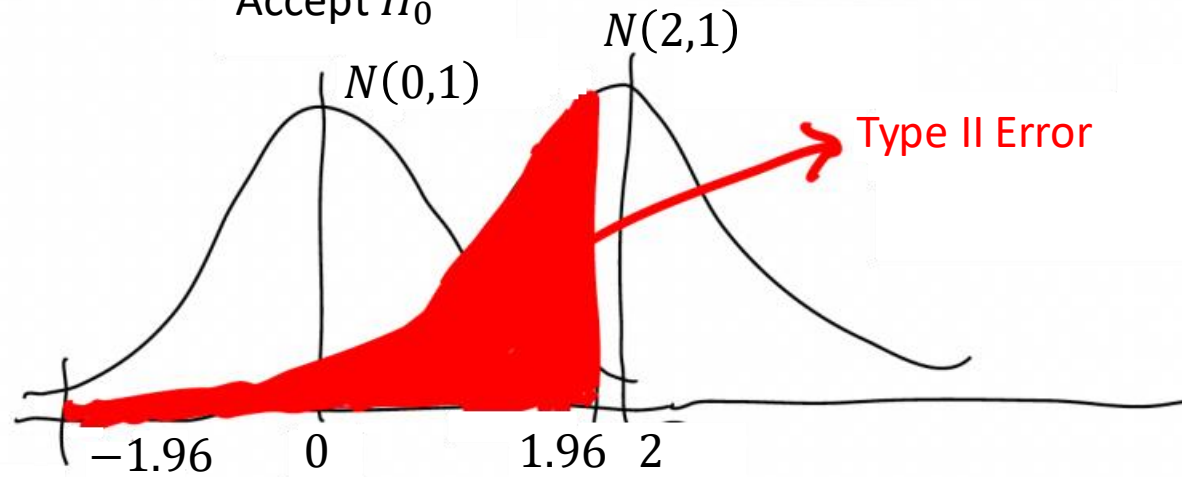


Figure 2

Question 4

Let $Y_1, \dots, Y_n$ be i.i.d. draws from a distribution with mean μ .

A test of $H_0: \mu = 10$ vs. $H_1: \mu \neq 10$ using the t-test statistic yields a p-value of 0.07.

- (a) Does the 90% confidence interval contain $\mu = 10$? Explain.
- (b) Can you determine if $H_0: \mu = 7$ is rejected by the (two-sided) t-test with 5% significance level? If yes, provide the reason. If no, provide the counter-example.

[Hint: Is the t-statistic for $H_0: \mu = 7$ on the left-hand-side or right-hand-side of the t-statistic for $H_0: \mu = 10$?]

Question 4

(a)

For $H_0: \mu = 10$, the p-value is 0.07, which is smaller than 0.1, so we can reject $H_0: \mu = 10$ at the 10% significance level. Since we can reject it at the 10% level, then $\mu = 10$ is not contained in the 90% confidence interval.

(b)

For testing $H_0: \mu = 7$, note that the corresponding t-statistic can be written as:

$$t_7 = \frac{\hat{\mu} - 7}{SE(\hat{\mu})} = \frac{\hat{\mu} - 10 + 10 - 7}{SE(\hat{\mu})} = \frac{\hat{\mu} - 10}{SE(\hat{\mu})} + \frac{3}{SE(\hat{\mu})} = t_{10} + \frac{3}{SE(\hat{\mu})} > t_{10}$$

which follows from the fact that $SE(\hat{\mu})$ is always positive.

Therefore, t_7 is always on the right-hand-side of t_{10} .

Question 4 (b)

Furthermore, we notice that for testing $H_0: \mu = 10$, the p-value is 0.07, so it cannot be rejected by a (two-sided) t-test at the 5% level. Therefore, t_{10} will be inside the acceptance region from -1.96 to 1.96 . On the other hand, whether t_7 is inside the region from -1.96 to 1.96 depends on the value of $SE(\hat{\mu})$.

Let us think about some examples. Suppose the standard error $SE(\hat{\mu})$ is very small such as 0.001, then $t_7 = t_{10} + 3000$, so that t_7 must be outside the acceptance region $[-1.96, 1.96]$. Then, suppose the standard error is very large such as 1000, then $t_7 = t_{10} + 0.003$, so that t_7 is very close to t_{10} . In this case, t_7 will be inside the acceptance region. In sum, depending on the value of the standard error, t_7 can be either inside or outside the acceptance region. So, we cannot determine if $H_0: \mu = 7$ is rejected at the 5% level by the information at hand.